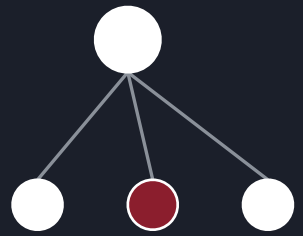
Sufficiently Advanced Agents Are Indistinguishable from Erlang:

An Operational Mapping from OTP to Multi-agent LLM Systems



Can OTP help AI systems succeed reliably despite component failures?

99.9999999%

service availability

Telecom achieved this a quarter century ago, with components engineers knew would fail.

British Telecom, 2002
Cronqvist, Erlang Factory 2010

Mick Darling, Dollhouse Research

MAS-GAIN ’26, 16 October 2026

Waiting without an active model call

Waiting belongs in ordinary code

- Some agents waited hours for outside events. Repeated polling calls consumed most of a weekly quota.
- Parking keeps the agent's identity and mailbox in the server.
- While parked, no model call is active and no inference tokens are spent.

Component	Responsibility
Model	Generates a response during a call
Harness / client	Calls the model and accesses MCP tools
MCP server	Retains agent identity, messages and waiting state

An event marks work ready. A client or dispatcher then arranges the next model call.

An idle session does not inherently consume inference tokens.

Workers and supervision

Worker

Managed work retains identity, state and a mailbox across model calls.

Supervisor

Ordinary code detects failures, replaces failed activations, limits restarts and escalates.

Date	Evidence
April 2026	Operational mapping recorded in dated engineering issues
July 2026	Preregistered experiments run

A mapping that makes testable predictions

Multi-agent LLM systems face the constraint set that produced Erlang/OTP, so OTP's constructs, discipline, and failure catalogs transfer testably

1

A graded mapping

Twenty OTP constructs against agent behavior, each row exact, loose, or breaking

2

Pre-registered experiments

Supervision architecture crossed with model tier, sealed and replayable

3

What does not transfer

Stated as the research agenda, not as footnotes

Twenty constructs, three grades

12 exact

Claim-relevant semantics preserved

- Lightweight process
- Registered names
- Global send
- Mailbox / selective receive
- Immutable messages / atomic receive
- Supervision strategies
- Restart intensity
- Behaviours / enforced interfaces
- DynamicSupervisor
- ETS shared read state
- Process groups
- Exit / suspend / resume

5 loose

Documented difference

- Per-process heap
- Hot code loading
- Let it crash
- Generational GC
- Mnesia transactions

3 breaking

No faithful analog

- Share-nothing default
- Typed term payloads
- Preemptive scheduling

Author-coded mapping from dated engineering issues. No independent audit.

Missing machinery and predicted failures

Gap in our runtime	Failure OTP predicts, and we saw	Standard OTP fix
Mailbox entries editable after append; consume not atomic	Tamper window	Append-only entries, content hash, atomic consume
No restart ceiling on supervisors	Runaway restart loop	Restart intensity limit with escalation
Worker lifetime not tied to the client session	Orphaned workers ran for days after sessions died	Lifecycle links
Single-waiter delivery	No fan-out	:pg process groups
Mailboxes serving read-heavy state	Poor fit, slow reads	ETS shared read store

April issues preceded matching incidents. The designers already knew Erlang.

Three limits of the mapping

Share-nothing default

Erlang
Private process heaps provide isolation by default.

Evaluated agent systems
Fresh contexts and access rules enforce isolation as policy.

Policy can be audited and subverted.

Typed term payloads

Erlang
Structured terms are matched as data.

Evaluated agent systems
A valid envelope can still contain prose that redirects the receiver.

Typed semantic layer: preliminary, unevaluated.

Preemptive scheduling

Erlang
The runtime can preempt a process at reduction boundaries.

Evaluated hosted interfaces
Timeouts and between-call control cannot pause and resume generation.

This is a limit of the interfaces evaluated in the study.

Experimental configurations

Standalone research harness.
Supervisors, validators and routing are ordinary code.

Arm	Configuration	Used in
A	Supervised, with fresh-context replacements	M1, M2, M2b
B	Fixed retry retaining prior context	M1
C	Bare loop with no retry	M1
S	Multiple candidates and a deterministic selector	M2, M2b

Sequential scheduling. LLM-worker counts only.
No concurrency, latency or throughput result.

Observed signals

Process failure, timeout, invalid messages, failed output checks and restart intensity.

Actions

Terminate, replace with fresh context, route, quarantine or escalate.

Blind spot

Wrong answers that pass configured checks raise no supervisor signal.

Tasks, models and success measures

Tier	Model
F2	Claude Sonnet 5
F1	Claude Haiku 4.5
T5	Llama-3.3-70B
T4	Qwen3-32B
T3	Qwen3-14B
T2	Qwen3-8B
T1	Ministral-3B
T0	Qwen3-1.7B

Provider and model version pinned for every episode.

Tasks

M1 and M2: GAIA-style multistep tasks with normalized exact-match scoring.

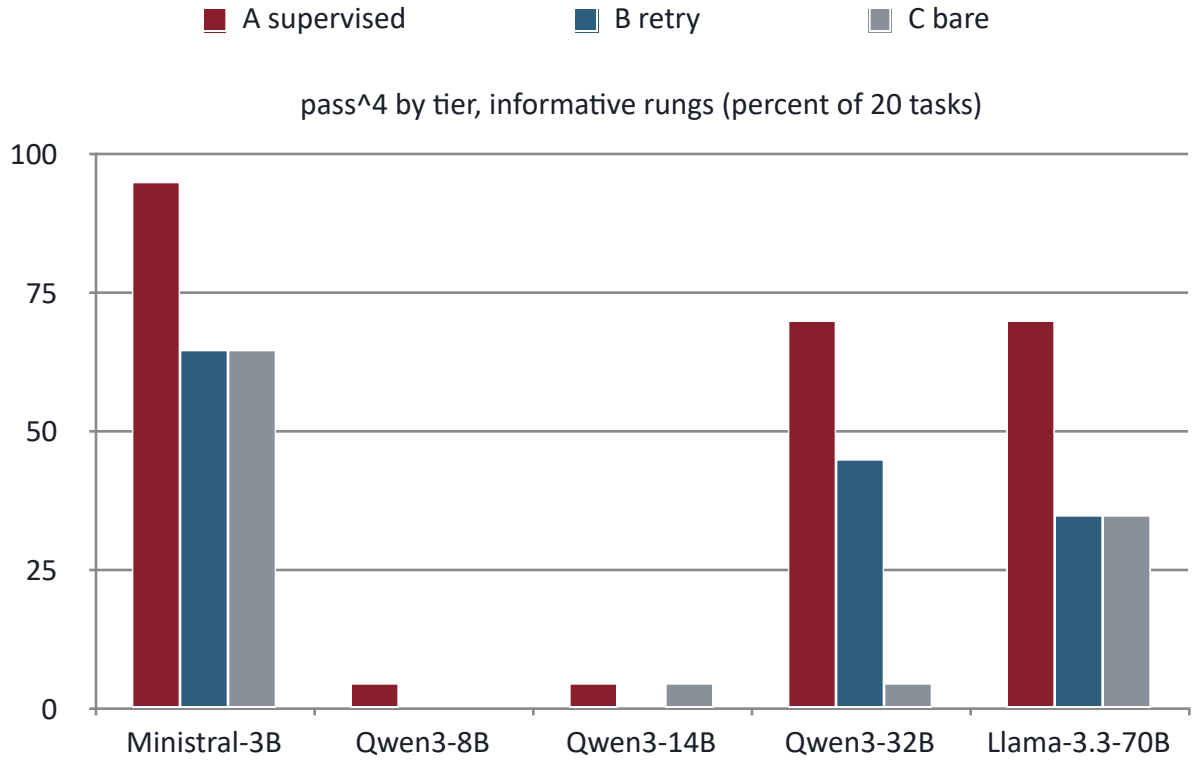
Scaling: proof graphs, provenance resolution, fault-hypothesis graphs and constrained allocation.

Three measures of success

M1: all four runs correct (pass^4)
M2/M2b: correct individual runs
Scaling: correct individual components

Retries stay inside a run. M1 analysis unit: task.
Rules freeze before inference. A missed gate withholds its conclusion.
Software-engineering proxies. End-to-end coding untested.

The architecture matters within a model tier



20 capability tasks per result
Each task must succeed on all four runs

32B: 70% supervised, 45% retry, 5% no retry

Frontier models reach 100% in every arm; the smallest reaches zero

Pilot resolution. No individual-result significance claim. Treatment combines several choices.

Supervision versus the sampling control

M2: 3B, 32B and 70B. M2b: fresh 32B runs.
Metric: correct individual runs. Paired analyses keep attempts grouped by task.

Study	Supervised	Best-of-N	Delta (pp)	95% CI (pp)	p-value
M2	198/240	118/240	+33	[22.5, 45.0]	1.9e-6
M2b	60/80	39/80	+26	[8.75, 42.5]	0.0035

0 additional successes over the first candidate across 320 sampling-control runs

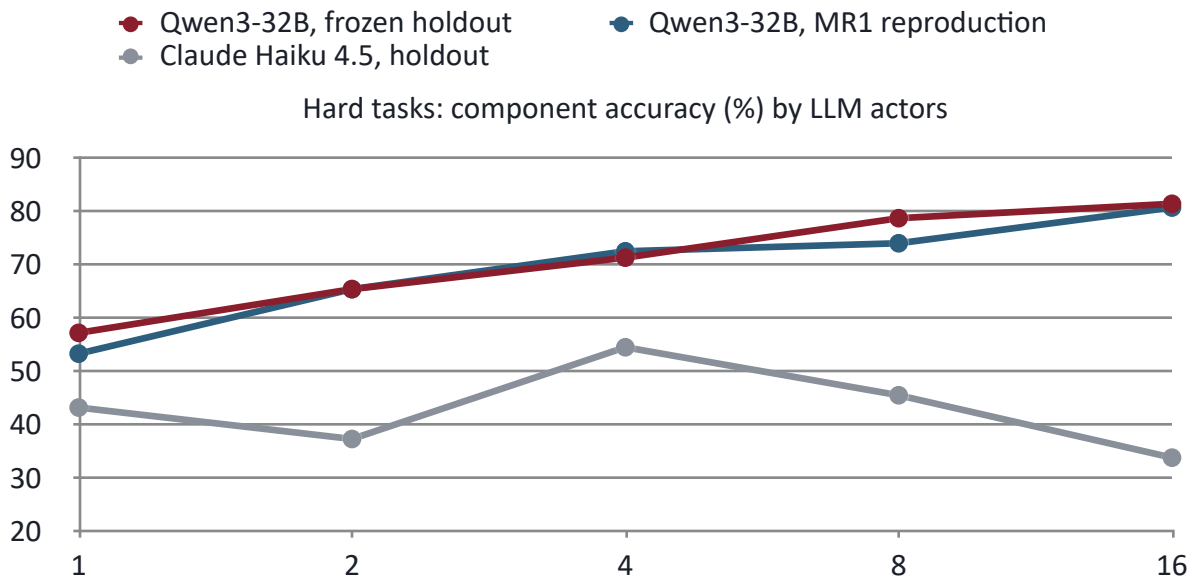
Compute parity missed

32B sampling/supervision token ratios:
M2: 0.836. M2b: 1.117.
Target band: 0.90 to 1.10.

This selector and schedule. No isolated-mechanism or matched-compute conclusion.

Decomposition depends on the model

The apparatus supplies the decomposition and routing plan.
Workers run sequentially. Accuracy counts individual components.



Sonnet 5 excluded: JSON batching artifact.

Qwen3-32B

57.4% to 81.6%
+24.2 percentage points
95% CI [13.3, 35.5]

Reproduced curve

53.5% to 80.9%
Matched-resource gate failed.

Haiku 4.5 peaks at 4

54.7% at 4 workers
34.0% at 16 workers

Open problems in the transfer

Semantic failure detection	A crashed process provides a clear failure signal.	An incorrect answer that passes configured validators can leave the supervisor with no recovery trigger.
Context leakage and isolation	Ordinary process memory is isolated by the Erlang runtime.	Fresh contexts and access rules limit influence. Data on an allowed path can still carry an injection attack.
Restart economics	Lightweight process restarts are cheap.	Restarts cost tokens, time and money. Their isolated benefit and matched-cost savings remain unestablished.

Recovery requires an observable failure and a budget that makes replacement worthwhile.

Mapping grades and evidence

An exact mapping describes an operational correspondence.
It does not certify implementation or isolate a mechanism's contribution.

Evidence level	Meaning	Example from the paper
Tested	Sealed, preregistered evidence	Integrated supervision and decomposition
Spot	Validation or exploratory evidence	18/18 deterministic injected-fault outcomes
Gap	Designed, with no experimental result	Continuous feedback control or heterogeneous fleets

Next tests for the mapping

The paper proposes these tests. They are not confirmatory results.

Question	Next experiment	Purpose
Injected faults	Supervision versus retry on a paid fault grid	Whether injected faults widen the supervision gap
Small-model parity	Supervised small model versus a defined frontier baseline	Whether a parity tier exists, with its cost ratio
Choosing actor count	Prospective sweeps across models and task families	Whether granularity signatures are predictable

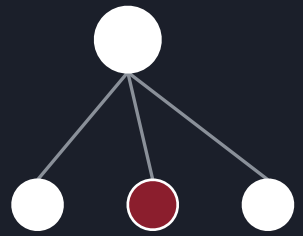
Each conclusion must meet its preregistered acceptance conditions.

What the experiments establish

The measured system includes the model, supervision, validators and retained task state.

Supervision	At 3B, 95% of tasks passed all four runs with supervision, versus 65% with no retry.
Supplied decomposition	Qwen3-32B component accuracy rose from 57.4% to 81.6% as worker count increased from 1 to 16.
Model dependence	Haiku peaked at 4 workers, then fell to 34.0% at 16. Worker count must suit the model and task.

Paper: doi.org/10.1145/3843778.3856135



Thank you

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Backup: How the mappings are graded

All 20 constructs from the paper. Grades describe the operational analogy.

Grade	Rows	Grading rule	Concrete example
Exact	12	Every semantic used by the comparison has a direct counterpart.	A parked agent keeps its identity and mailbox without an active model call.
Loose	5	The function transfers, with a documented material difference.	Clean restart is possible, but a wrong answer may emit no failure signal.
Breaking	3	The evaluated substrate offers no faithful counterpart to the guarantee.	A timeout does not provide pause and resume inside hosted inference.

An exact analogy can still be an implementation gap in the paper's runtime.
These author-coded judgments have not had an independent audit.

Paper §2, Table 1 and the auxiliary grading rule. Evidence identifiers M01–M20 follow the final table order.

Exact: Parked actors and identity

Exact mappings M01–M02. Continuing identity is separate from an active model call.

OTP contract	Agent counterpart	Why this grade / boundary
M01 Lightweight process, free when idle **Identity and mailbox persist while blocked.**	An agent waits with an addressable identity and server-side mailbox, with no resident or outstanding model activation.	Waiting and addressability survive without inference-token consumption. Storage and wake-up costs still exist.
M02 Registered name, :global **A stable name resolves to a live entity.**	A registrar maps an agent name to its current runtime identity.	Stable lookup is preserved. Registry scope and distributed consistency remain implementation details.

An agent can continue to exist while the model is idle or unavailable.
A durable identity lets work resume when the next event arrives.

Paper §2.1, Table 1. Evidence key M01–M02. Analogy grade; implementation coverage varies.

Exact: Delivery and mailboxes

Exact mappings M03–M04. Routing and selection belong to the runtime.

OTP contract	Agent counterpart	Why this grade / boundary
M03 :global.send **Resolve a name, then deliver the message.**	The runtime resolves the named recipient and routes the message into its mailbox.	Name resolution followed by direct delivery is preserved. Transport and persistence may differ.
M04 Mailbox, selective receive **Queue messages and select a matching shape.**	A per-agent mailbox uses typed filters to select the matching queued message.	Queuing and selective delivery transfer. A type filter does not establish the safety of prose content.

The model does not need to poll or interpret every message to decide where it belongs.
Runtime routing and filtering can determine which activation receives which input.

Paper §2.1, Table 1. Evidence key M03–M04. Analogy grade; implementation coverage varies.

Exact: Integrity and supervision

Exact mappings M05–M06. Message integrity and restart topology are explicit contracts.

OTP contract	Agent counterpart	Why this grade / boundary
M05 Message immutability, atomic receive **Immutable message, one atomic removal.**	Append-only, content-hashed entries and an atomic consume operation.	The message cannot change during consumption, and one selected entry has one atomic removal. Durability is additional.
M06 Supervision strategies **One-for-one / one-for-all**	Supervisor policy restarts either one child or the supervised group, using clean context.	Restart topology transfers directly. Detecting a semantically wrong answer remains a separate problem.

The runtime can make message consumption and restart policy deterministic.
Whether a model answer counts as a failure remains a separate detection problem.

Paper §2.1–2.2 and 3, Table 1. Evidence key M05–M06. Analogy grade; implementation coverage varies.

Exact: Limits and behaviours

Exact mappings M07–M08. The grade assumes a rule the runtime actually enforces.

OTP contract	Agent counterpart	Why this grade / boundary
M07 Restart intensity limits **Bound restarts in a time window.**	A restart ceiling stops or escalates repeated worker failure.	The bounded-restart rule is the same. Model-call budgets differ. The ceiling was not independently ablated.
M08 Behaviours: gen_server, supervisor **A role defines callbacks and a runtime protocol.**	An agent role has a fixed interface enforced by runtime code.	Exact when the interface is enforced. Prompt-only roles fall short; the paper lists enforcement as a runtime gap.

A role description becomes an engineering contract only when the runtime can enforce it.
Restart ceilings bound recovery without proving that recovery will succeed.

Paper §2.1, 3 and 6, Table 1. Evidence key M07–M08. Analogy grade; implementation coverage varies.

Exact: Dynamic workers and shared state

Exact mappings M09–M10. Worker creation and shared reads sit outside model context.

OTP contract	Agent counterpart	Why this grade / boundary
M09 DynamicSupervisor **Start child workers on demand.**	A supervisor creates workers from a template and owns their lifecycle.	On-demand child creation and ownership transfer. Experimental fan-out is a bounded special case.
M10 ETS shared read state **Share a table outside private actor state.**	A supervised external store supplies state optimized for shared reads.	Explicit shared state outside actor context is the same architectural escape hatch. Store details vary.

Workers can read explicit shared state without carrying the entire history in every context.
The shared-store analogy does not require uncontrolled sharing between all actors.

Paper §2.1, 3 and 6, Table 1. Evidence key M09–M10. Analogy grade; implementation coverage varies.

Exact: Groups and lifecycle control

Exact mappings M11–M12. Lifecycle control here stops at activation boundaries.

OTP contract	Agent counterpart	Why this grade / boundary
M11 :pg process groups Named membership supports group delivery.	A runtime tracks named groups and delivers across current agent membership.	Membership-scoped delivery transfers. The paper lists production fan-out as designed, not experimentally established.
M12 exit/2, suspend, resume Control a target outside ordinary messages.	The control plane kills, pauses or resumes agent work between activations.	Exact only at activation boundaries. Full control-plane support is a runtime gap; mid-inference control is M20.

Group membership and lifecycle control can be runtime responsibilities.
Control between activations does not supply preemption inside a running inference call.

Paper §2.1 and 6, Table 1. Evidence key M11–M12. Analogy grade; implementation coverage varies.

Loose: Isolation and instruction swaps

Loose mappings M13–M14. The useful function survives, with a weaker or incomplete contract.

OTP contract	Agent counterpart	Why this grade / boundary
M13 Per-process heap **Private memory blocks direct cross-process references.**	The supervisor creates fresh contexts and passes earlier state explicitly.	Isolation is a runtime policy. It lacks the same substrate guarantee and can leak through shared history or prose.
M14 Hot code loading, code_change **Upgrade code and convert existing state.**	Instructions can change while an agent is parked.	Instruction replacement transfers. There is no standard migration callback for in-flight agent state.

Fresh contexts and parked-agent upgrades are useful engineering mechanisms.
Their correctness depends on policy and state handling that the runtime must supply.

Paper §2.2–2.3 and 6, Table 1. Evidence key M13–M14. Analogy grade; implementation coverage varies.

Loose: Let it crash

Loose mapping M15. Recovery transfers, while the failure signal and restart economics change.

Aspect	OTP contract	Agent counterpart	Material difference
Recovery	A failed worker terminates and a supervisor replaces it.	Kill a derailed activation and restart with clean context.	A clean replacement is available once the system recognizes failure.
Failure signal	A worker termination gives the supervisor a failure signal.	A wrong answer can be confident and well-formed.	A validator must detect the problem. Valid format alone does not prove correctness.
Recovery cost	Restart is cheap enough to be a routine strategy.	A new activation spends tokens and takes seconds.	The system needs a bounded recovery budget and an escalation rule.

The restart action transfers, while silent errors and token costs change the contract.
Recovery therefore depends on validators and an explicit restart budget.

Paper §2.2 and §3. Evidence key M15. Restart topology is exact in M06; the broader failure strategy is loose.

Loose: Compaction and transactions

Loose mappings M16–M17. Similar purposes do not imply the same correctness guarantees.

OTP contract	Agent counterpart	Why this grade / boundary
M16 Generational garbage collection **Reclaim unreachable memory, retain live values.**	Context compaction summarizes or deletes earlier text to manage a bounded window.	Compaction can lose relevant meaning. It does not trace reachability or preserve exact values. Illustrative only.
M17 Mnesia, distributed transactions **Atomic updates across replicated records.**	Saga-style compensation repairs partial effects at application level.	Compensation does not provide distributed ACID. The paper claims no Dollhouse production transaction result.

Both analogies address familiar resource or coordination problems with weaker contracts.
Compaction can lose information, and compensation can leave partial effects to repair.

Paper §2.2 and 6, Table 1. Evidence key M16–M17. Analogy grade; implementation coverage varies.

Breaking: Isolation by default

Breaking mapping M18. Boundary of the hosted-agent setting evaluated in the paper.

OTP guarantee

Ordinary process state is private by default.
Another process cannot directly reference or mutate that heap state.

Evaluated agent boundary

Accumulated context and shared histories permit influence across agent work.
Fresh-context isolation requires an enforced policy.

Illustration

Appending one agent's output to a shared history makes it available to later activations.
A fresh-context policy can control that flow, but the substrate does not supply the same default.

M13 is loose because an isolation policy can perform the useful function.
M18 is breaking because the substrate does not supply the same hard guarantee.

Paper §2.3. Evidence key M18, compared with M13. The claim concerns the evaluated substrate, not every agent system.

Breaking: Payload interpretation

Breaking mapping M19. Boundary of the hosted-agent setting evaluated in the paper.

OTP guarantee

Receivers match structured terms against expected shapes.
The payload does not automatically become a natural-language instruction.

Evaluated agent boundary

A model interprets prose in a message.
Content can redirect its behavior even inside a correctly typed envelope.

Illustration

A result message can have a valid type while its text asks the receiver to ignore the task.
Transport validation can accept the shape without establishing safe interpretation of the content.

A typed mailbox can decide which message to deliver, which is exact in M04.
It cannot supply typed semantics for model-interpreted prose, the break in M19.

Paper §2.3. Evidence key M19, compared with M04. The proposed semantic instruction layer remains unevaluated.

Breaking: Preemption during inference

Breaking mapping M20. Boundary of the hosted-agent setting evaluated in the paper.

OTP guarantee

The scheduler can interrupt processes at reduction boundaries.
Runnable work can progress without application cooperation.

Evaluated agent boundary

The evaluated hosted interfaces expose monitors and timeouts.
They do not expose suspend and resume of internal generation state.

Illustration

A timeout lets a caller stop waiting; it does not establish that provider inference paused.
It also provides no way to resume the same internal generation state.

M12 covers control between activations; M20 covers control inside a model call.
This limit concerns the evaluated interfaces. Other serving stacks may offer stronger control.

Paper §2.3 and Table 1. Evidence key M20, compared with M12. Scope: the hosted interfaces evaluated for the paper.

Backup: sampling-control details

Metric	M2	M2b replication
Model tiers	3B, 32B, 70B	32B
Episodes per arm	240	80
Correct supervised runs (A)	198/240	60/80
Correct best-of-N runs (S)	118/240	39/80
Paired difference (percentage points)	+33	+26
95% CI (percentage points)	[22.5, 45.0]	[8.75, 42.5]
Wilcoxon p-value	1.9e-6	0.0035
Sampling / supervision tokens at 32B	0.836	1.117
Selection gain over the first sample	0	0
Verdict	Directional	Directional

Selector: deterministic, zero-token, preregistered. Parses the output contract, uses canonical majority vote when multiple candidates fit the budget, otherwise the earliest valid answer. It sees neither gold answers nor graders.

Selection gain counts added successes over the first sample. Both studies missed the 10% token-parity band in opposite directions. M2b used held-out, never-scored attempts.

Backup: related work

- Actors and OTP: Hewitt 1973; Armstrong 2003. Conscious adopters: AutoGen's event-driven redesign, Jido on the BEAM, Akka's agentic platform. No novelty claim about conscious adoption; our subject is runtimes that converge without choosing it
- Failure taxonomies: MAST (NeurIPS 2025), most multi-agent failures stem from system design and orchestration; MAS-FIRE, supervised closed-loop topologies neutralize faults that collapse pipelines
- Selection without verification: CodeMonkeys loses gains at the selection step at frontier scale; we measured zero selection gain at small-model scale
- AgentRadio: over ten points from passive step-boundary peer-message delivery, published eight days after our submission
- Runtime patterns: Srinivasan 2026 places a supervisor between LLM output and the world; Guimaraes 2026 argues agent frameworks are re-deriving BEAM infrastructure
- Evaluation discipline: tau-bench pass^k, GAIA tasks, Kapoor et al. strong simple baselines, Miller error bars, METR horizons

Manifests, preregistration digests and replay tool on request. AGPL apparatus release intended.

Backup: GAIA-style task suite

Controlled tool tasks used in M1 and the M2/M2b comparisons.
Each scored bundle has two independent steps and returns JSON scalar answers.

Difficulty	Tasks	Required work in the scored M1 suite
Easy	7	Look up an approved value in pinned evidence.
Medium	7	Read a local table, then calculate a sum or product.
Hard	6	Combine two evidence lookups with arithmetic.

Scoring and study design

- 20 scored tasks. M1: 3 arms, 8 tiers, 25 bundles and 4 attempts = 2,400 episodes.
- 5 bridge bundles support failure analysis and are excluded from capability scoring.
- Answers are normalized, then graded by exact match.
- M1 pass^4 requires all four attempts correct. M2/M2b score individual runs.

Scope: a controlled GAIA-style suite. Results are not full GAIA benchmark scores.

Backup: Budget-governed uplift

Beyond the paper. Not peer reviewed. Review welcome.

Qwen3-32B through the pinned OpenRouter / SiliconFlow FP8 route.
96 episodes. Each regime / actor cell has 16 task-attempt clusters and 256 components.

Shared entitlement	N=1	N=4	N=16	N16–N1	95% cluster bootstrap
60,000 tokens	25.39%	28.91%	31.25%	+5.86 pp	+1.17 to +10.94 pp
$0.015 provider cost	25.00%	30.47%	32.42%	+7.42 pp	+3.13 to +12.11 pp

OTP improved component accuracy under both common token and cost caps.
The finding supports coordinated actors within enforced resource budgets.

Both preregistered co-primary contrasts passed. N=16 used more tokens and provider spend.
This route-specific result closes the matched-resource confirmation gap.
Provider invariance remains untested. The paper's historical MR1 economics remain withheld.

SiliconFlow budget-governed experiment. Run: 25 July 2026. Component accuracy, not M1 pass[4].

Backup: Eight-actor operating envelope

Beyond the paper. Not peer reviewed. Review welcome.

Component accuracy: 320 episodes, concurrent Qwen3-32B actors, pinned SiliconFlow route.

Actors	Unconstrained	60-second cap	60,000-token cap	$0.015 provider cap
1	26.95%	25.39%	24.22%	25.00%
2	26.56%	28.13%	27.73%	30.08%
4	29.69%	31.25%	31.25%	31.64%
8	**35.16%**	**35.94%**	**25.00%**	**35.94%**
16	32.42%	32.81%	0.00%*	0.00%*

Eight actors balanced quality and latency in the unconstrained and 60-second regimes.
The drop at sixteen shows why actor count must be checked against resource limits.

N=8 vs N=1, unconstrained: +8.20 pp, 60.28% less wall time, 17.07% less cost per correct component, with 8.17% more cost per episode. At 60 seconds: +10.55 pp. This is specific to this setup.
*N=16 hit token/cost admission limits. Lower bills reflect incomplete work. Costs are partial.

Actor-count operating-envelope experiment. Run: 25 July 2026. Costs include provider plus active runner only.

Backup: Luna holdout

Beyond the paper. Not peer reviewed. Review welcome.

D1 = direct. G# = # actors. Each arm: 12 clusters, 192 components. 384 calls overall.

Arm	Correct	Accuracy	vs D1
D1 direct	156/192	81.25%	—
G1	148/192	77.08%	−4.17 pp
G2	152/192	79.17%	−2.08 pp
G4	**167/192**	**86.98%**	**+5.73 pp**
G8	154/192	80.21%	−1.04 pp
G16	165/192	85.94%	+4.69 pp

The G4 result needs replication across more tasks and fresh runs.

G4 paired 95% interval:
−2.30 to +13.75 pp
Holm-adjusted p = 1.0

10 of 11 net additional correct components came from 2 clusters.
One stochastic run per task / arm.

No OTP-versus-D1 contrast was significant. Half of direct components were at ceiling.
Concurrency was 1, with zero retries or restarts. This measures sharding and assembly.

Luna Level-2 holdout. Run: 3 August 2026. gpt-5.6-luna via direct OpenAI Responses.

Backup: Qwen3.8 Max ceiling

Beyond the paper. Not peer reviewed. Review welcome.

128 / 128 components correct

Direct baseline only. Eight calls across four families. Every family scored 32/32.

Its perfect baseline leaves no measurable headroom for OTP uplift.
A harder task set and verified reasoning/output limits are required.

Response budget	Recorded observation
Intended combined ceiling	4,096 reasoning + answer tokens per call
Observed completion usage	5,848–16,345 tokens per call
Scientific status	Intended response-budget treatment invalid

The score and route identity remain valid ceiling observations.
No overall authorized spend overrun occurred. This is not a controlled 4,096-token comparison.

Qwen3.8 Max Level-2 baseline. Run: 3 August 2026. Exact OpenRouter / Alibaba route.

Backup: DeepSeek Level 3 ceiling

Beyond the paper. Not peer reviewed. Review welcome.

Pinned OpenRouter / DeepSeek routes. Direct D1 qualification with high reasoning effort.
Each model: 8 calls, 128 components, four families, one response per task.

Model	Correct	Provider cost	Fixed wall time
DeepSeek v4 Flash	128/128	$0.091313	720.64 s
DeepSeek v4 Pro	128/128	$0.314855	1,413.80 s

Flash matched Pro's accuracy with less provider spend and elapsed time.
That makes Flash the cheaper first candidate for harder-task calibration.

Pro: 3.4481× provider cost, 1.9619× fixed wall. Neither model qualified for an OTP curve.
Earlier missing final answers at smaller budgets were response-budget diagnostics.
Times are from separate sequential runs. Direct DeepSeek API cells remain unrun.

DeepSeek Level-3 semantic evaluation. Run: 5 August 2026. Historical measured costs and times.

Backup: Deterministic output actors

Beyond the paper. Not peer reviewed. Review welcome.

Six new tasks across two model routes. One synthesis response per cell.
A deterministic actor knows the exact requested output field before inference.

Model route	Baseline	Request-bound actor	Rescues / harms
Qwen3-1.7B, HF / Featherless	3/6	6/6	3 / 0
Qwen3.6-27B, direct Groq	3/6	6/6	3 / 0
Combined	**6/12**	**12/12**	**6 / 0**

A deterministic actor rescued correct outputs by enforcing a known contract.
This is a system-level capability gain with no extra model call.

A key-agnostic alternative scored 12/12 live but admitted 2 unsafe adversarial fields and failed the stop rule.
The accepted actor uses a field declared before inference.
This small spot does not establish semantic field discovery or model-intelligence gains.

Request-bound output-schema spot check. Run: 22 July 2026. Exploratory whole-system result.

Backup: Gateway integration

Beyond the paper. Not peer reviewed. Review welcome.

Profile	What passed	Claim boundary
Real provider, 7 October	Codex tool turn and continuation. 2 OpenAI calls, 8 seconds.	Transparent forwarding. No semantic recovery test.
Synthetic provider, 9 October	Claude Code tool-input violation, restart and checked release. 3 synthetic posts, 0 paid calls.	Closed provider-free profile. No real-provider supervised accuracy result.

The gateway carried a native tool workflow through the tested route.
Supervised recovery still needs a controlled test with a real provider.

Gateway accuracy gains, native private-CA TLS and tool-consumption deduplication remain unqualified. Interrupted native consumption is unknown. The tests do not authorize public activation.

Gateway qualification snapshots, 7–9 October 2026. Distinct profiles with distinct evidence limits.

Backup: Replication and review

Beyond the paper. Not peer reviewed. Review welcome.

Next experiment: informative headroom

Calibrate each model and task family using direct-model outcomes only.
Target 40–80% baseline accuracy, preferably 50–70%. Reserve a disjoint sealed holdout.
Freeze selection before opening OTP outcomes, then compare the full actor-count curve.

Broader claims need tasks where the direct model has room to improve.
Prospective holdouts and repeated runs can test whether the gains generalize.

Interested reviewers are welcome

We welcome review of the methods, reports, verdicts and aggregate analysis in the handoff.
Full raw shards, transcripts and replay bundles are absent. Some experiments predate camera-ready.

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Evidence snapshot: 15 October 2026. Midpoint bracketing is a plan with no completed efficiency result.